Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

SDC HACKATHON 2026

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1. Team & Project Overview

Team Name	505 Server Error
Project / Solution Name	NGate
Name (Leader)	Melissa Ting Jia Jie
Email & Phone (Leader)	mtjiajie21@gmail.com & 010-9665679
University Affiliation(s)	Universiti Malaysia Perlis (UniMAP)
Live Prototype URL	https://n-gate-pos-v-5.replit.app/ngate/
Code Repository URL (public / access granted to judges)	https://github.com/jiaqi685/505-Server-Error

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

NGate is an AI-assisted procurement and inventory prototype. The React/Vite interface connects to a TypeScript/Express API hosted on Replit. Express manages Replit sign-in, sessions, quotation uploads and a protected read-only connection to a Python Flask service. PostgreSQL stores user and session records and upload metadata; private object storage holds quotation files. Google Gemini extracts structured fields from uploaded quotations, with missing or uncertain values returned for human review. Flask and SQLite support a team-created 22-product Krukkruk Mini Mart demo, stock indicators, quotation comparisons, purchase-order approvals and downloadable PDFs. The main React presentation currently uses mock data rather than live Flask records. Supplier messages and purchase orders are prepared for manual sending, not automatically transmitted. The owner controls supplier selection and approval. Appendix A illustrates the proposed POS-connected workflow; live POS synchronisation is not part of the current prototype.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
React + Vite	Browser interface and build tools	MIT
TypeScript ; Node.js + Express	Frontend-facing server, routing and sessions	Apache-2.0 ; MIT
Phyton + Flask	Inventory and purchasing demonstration API	PSF License; BSD-3-Clause
Drizzle ORM	PostgreSQL access	Apache-2.0
ReportLab	Purchase-order PDF generation	BSD-style license
SQLite + PostgreSQL	Demo inventory database; user/session/upload metadata	Public Domain; PostgreSQL License

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Google Gemini (via Replit integration)	Extract quotation details from uploaded JPG/PDF files	Replit-managed integration
Replit OpenID Connect	Authenticated user sign-in	Sponsor-provided

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit Cloud Hosting	Hosts NGate frontend, Express API and Flask backend	Sponsor-provided access / Region not specified
Replit PostgreSQL	Stores user accounts, login sessions and quotation upload metadata	Replit-managed database / Region not specified
Replit Private Object Storage	Securely stores uploaded quotation PDFs and images	Replit platform / Region not specified
Replit Authentication	Provides OpenID Connect user sign-in	Replit platform / Region not specified

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Google Gemini	Google, integrated through Replit	Pre-trained (AI model for extracting quotation details)	Google Gemini API Terms

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
NGate Synthetic Supplier Quotation Dataset (22 Products)	Team-created supplier quotation PDFs for Krukkruk Mini Mart demonstration	Synthetic	Team-owned; not externally licensed	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Replit AI	Code development, debugging, and analysis	Major
ChatGPT (OpenAI)	Technical writing and documentation	Moderate

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Our team maintained full responsibility for AI-assisted development and system outputs. AI-generated code was reviewed against NGate's requirements, while quotation extraction results were validated through structured data checks and user confirmation. Supplier comparisons and purchasing calculations use deterministic backend logic to ensure transparent, reproducible results. Missing or uncertain quotation details are flagged rather than assumed. Purchase orders require explicit owner approval, preventing autonomous purchasing decisions. The team remains accountable for system functionality, data accuracy, security, and final decisions.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit (SDC Sponsor)	Cloud Development & Hosting Platform	Used to develop, test and host NGate's React frontend, Express and Flask backend, database integrations, and quotation-processing workflow.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit was essential to the development, testing, and deployment of NGate's current prototype. It provides an integrated environment for hosting the React frontend, Express and Flask backend services, and supporting database and storage connections. However, NGate is not permanently dependent on Replit. Its modular architecture allows migration to alternative cloud platforms with appropriate configuration changes. This demonstrates technical flexibility, reduces long-term vendor dependency, and supports future scalability beyond the hackathon. Replit accelerated prototype development, while NGate's core business logic remains independently developed and portable.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

NGate processes user authentication details, session records, supplier quotations, and purchasing information. PostgreSQL stores user accounts, sessions, and document metadata, while uploaded quotations are maintained in private Replit object storage.

Data protection measures include OpenID Connect authentication, session-based access control, ownership verification, short-lived signed upload URLs, and file validation (PDF/JPG/JPEG, maximum 8 MB). Quotation documents are transmitted to Google Gemini only when users request AI extraction, with extracted information subject to human review.

The current demonstration uses synthetic business data rather than real customer transactions. A formal data retention and deletion policy will be established before production deployment.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

NGate's key technical considerations are data reliability, integration readiness, and third-party AI service availability.

The current prototype uses synthetic inventory data; live stock monitoring will require POS integration and reliable transaction records. Supplier lead-time estimates will also benefit from sufficient historical purchasing and receiving data.

To ensure reliable decisions, NGate validates AI-extracted quotation details, flags uncertain information, and applies deterministic calculations for supplier comparisons. Purchasing decisions remain subject to explicit owner approval. Incomplete or ambiguous quotation details require user verification before being used in purchasing calculations.

Google Gemini service availability and usage limits may affect quotation processing. Future development will focus on POS synchronisation, automated data integration, and operational monitoring to support scalable deployment.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

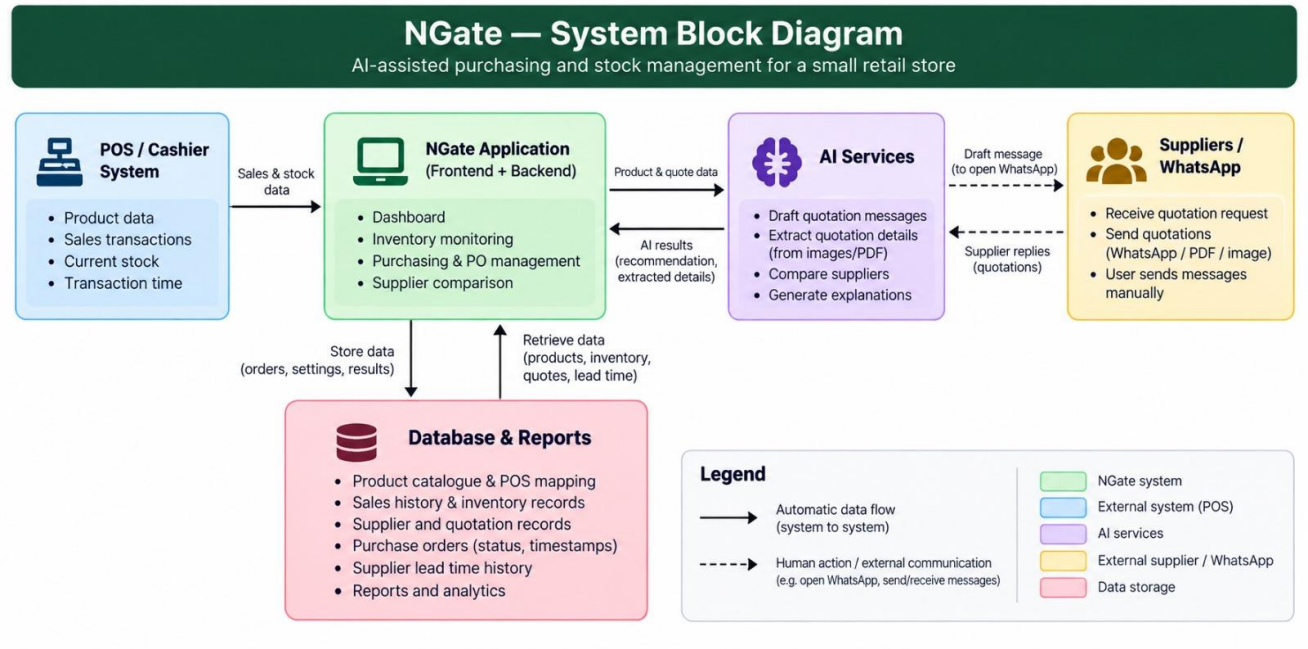
8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



24 OCTOBER 2026

Appendix A – NGate System Architecture and POS Integration Plan



Note: POS integration and automated supplier lead-time tracking are planned enhancements.